src\DINO\Dino_Game_basic.ino

```
1
2
3  #include <Servo.h>
4
5  Servo myservo;
6
7  // Servo control variables
8  const int SERVO_START_POS = 10;
9  const int SERVO_ACTUATE_POS = 0;
10 const int SERVO_PIN = 9;
11 const int JUMP_DELAY_MS = 580;
12
13 // LDR variables
14 const int LDR_PIN = A0;
15 const float LDR_LIGHT_THRESHOLD = 410.0;
16 const float LDR_DARK_THRESHOLD = 460.0;
17 float ldrValue = 0.0;
18
19 // Game state variables
20 int jumpCount = 0;
21 float DELAY_FACTOR = 0.66; // Factor to decrease delay as score increases
22 bool shouldJump = false;
23 bool isDay = true;
24 const int SERIAL_BAUD = 9600;
25 const unsigned long FIRST_CACTUS_JUMP = 1;
26 unsigned long firstCactusTime = 0;
27 bool cactusDetected = false;
28 int score = 0;
29 const int SCENE_CHANGE_SCORE = 700;
30 const int SCENE_CHANGE_DELAY_MS = 500;
31
32
33 void setup() {
34     Serial.begin(SERIAL_BAUD);
35     myservo.attach(SERVO_PIN);
36     myservo.write(SERVO_START_POS);
37 }
38
39
40
41 void tap() {
42     myservo.write(SERVO_ACTUATE_POS);
43     delay(200);
44     myservo.write(SERVO_START_POS);
45 }
46
47
```

```
48 void loop() {
49     Serial.print("Score: ");
50     Serial.println(score);
51 
52     // Calculate score based on time since first cactus
53     if (cactusDetected) {
54         unsigned long elapsedTime = millis() - firstCactusTime + 35;
55         score = elapsedTime / 100;
56     }
57 
58     // Change scene every SCENE_CHANGE_SCORE points
59     if (score % SCENE_CHANGE_SCORE == 0 && score > 0) {
60         delay(SCENE_CHANGE_DELAY_MS);
61         isDay = !isDay;
62         Serial.println("Scene changed");
63     }
64 
65     // Gather and average LDR measurements
66     float sum = 0.0;
67     for (int i = 0; i < 10; i++) {
68         sum += analogRead(LDR_PIN);
69     }
70     ldrValue = sum / 10.0;
71     Serial.print("Average light intensity: ");
72     Serial.println(ldrValue);
73 
74     // Determine if jump is needed based on scene
75     shouldJump = isDay ? (ldrValue >= LDR_DARK_THRESHOLD) : (ldrValue <=
   LDR_LIGHT_THRESHOLD);
76 
77     if (shouldJump) {
78         Serial.println("Jumping!");
79         delay(JUMP_DELAY_MS - score * DELAY_FACTOR);  // Decrease delay as score increases
80         tap();
81         jumpCount++;
82 
83         // Record time of first cactus detection
84         if (!cactusDetected && jumpCount >= FIRST_CACTUS_JUMP) {
85             firstCactusTime = millis() - 35;
86             cactusDetected = true;
87             Serial.print("First cactus detected, beginning timing: ");
88             Serial.println(firstCactusTime);
89         }
90     }
91 }
92 
93 
```